

A reverse audit from claim to signal

Facilitation plan, reference reasoning, and review criteria

Teaching purpose. The goal is a justified decision to retain, repair, or withhold a visualization claim. Three complementary activities jointly examine four layers—graphical encoding, feature semantics, temporal support, and observation support. Feature and temporal-support reasoning are developed together; reference calculations support the audit decisions.

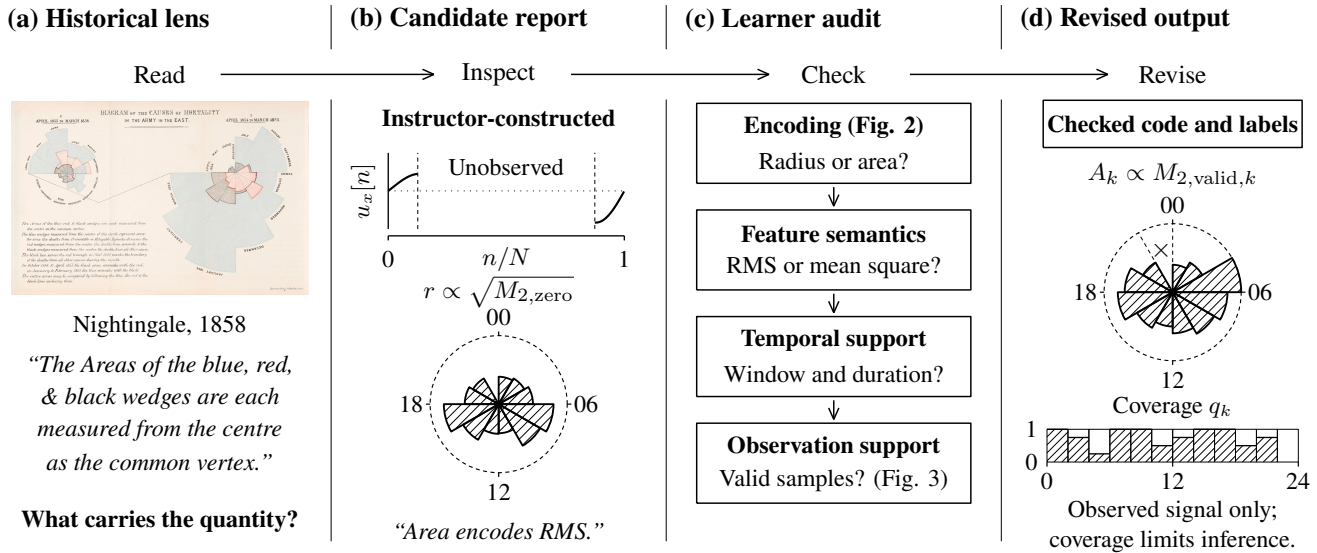


Figure 1 contains worked answers. Use it for synthesis or instructor preparation, not as the first student handout. The historical image is from Wellcome Collection, Public Domain Mark; modern plots are synthetic.

A suggested 90-minute session

Minutes	Activity	Proposed facilitation
10	Read (Card 0)	Identify the geometric carrier; separate reading instructions from the chart's silhouette.
15	Inspect (Card 1)	Record an individual provisional claim, ratio rationale, and information request.
40	Check (Cards 2–4)	Release geometry, feature/code, and coverage in order; use paired review with individual notes.
20	Revise (Card 5)	Submit code/annotations, figure, caption, and evidence-linked decision trail.
5	Debrief	Revisit the historical instruction and the limit of what the report establishes.

Adjust the timing to the class and available teaching time. A print-led class can omit execution; a programming laboratory can require code revision.

Facilitation and reference decisions

Alternative defensible paths matter more than a fixed number of errors

1. Read a difference

Ask for the object underlying the ratio, not only the ratio itself. The unannotated pair on Card 1 uses equal angles and radii in the ratio 1:2. Before the encoding is stated, geometry alone does not identify which data quantity is intended: twofold is a length reading, fourfold an area reading. After Card 2 specifies area and values 25 and 100, candidate A is consistent; candidate B produces a 1:16 area ratio and conflicts with the same declaration.

A response that gives fourfold without naming area needs a different follow-up from a response that incorrectly calculates area. Ask “What did you measure?” rather than assume a particular visual bias. Retain a correct initial answer with justification.

2. Feature and temporal-support audit

Candidate C is consistent: radius linear in RMS produces area linear in mean square. This prevents a rule of blanket rejection. In the candidate report from Card 1, area encodes zero-filled mean square but is labeled RMS. If mean square is intended, changing the label repairs that inconsistency. If area should encode RMS, recompute the radial mapping instead. Keep the coverage question open in both cases.

An answer identifies the feature, units, geometric carrier, and time window. Whole-window RMS cannot locate a burst; equal RMS does not imply equal duration-weighted accumulation. “Looks misleading” does not locate a repair. Neither the label “activity” nor a statistically correct feature establishes fatigue or mechanical power. These synthetic values use arbitrary acceleration units, not calibrated physical units.

3. Observation-support audit

D and E use the same observed squared sum. Their 25-to-100 difference comes from dividing by 800 versus 200 positions, not from two people's movement. E describes the observed-sample statistic but is not full-window ground truth. D's fatigue statement is unsupported. For no valid observations, retain an undefined feature (for example, NaN), show a missing marker, and report zero coverage.

A supported revision is: “The sectors summarize observed mean-square acceleration; incomplete coverage limits inference about the full intervals.” A stronger statement requires additional evidence, not simply a new caption or a different plot. Ask learners which claims they retained, qualified, or withdrew and what caused each decision.

The combined case in Fig. 1

The candidate report in Fig. 1(b) is the report introduced on Card 1 and revisited on Card 3. It is instructor-constructed, not a record of an actual model response. It combines a feature-label inconsistency with a missing-data decision. The revision in Fig. 1(d) uses the same synthetic record and scale, identifies the observed statistic, and reports coverage. It does not make time support and valid-sample support interchangeable. “Checked code and labels” names a learner deliverable: a computation and declaration checked against the stated feature, scale, and validity rule.

Reference calculations

Technical support for the geometry, feature, and observation-support audits

Geometry and feature semantics

For angle $\Delta\theta$ in radians and zero inner radius,

$$A = \frac{1}{2}\Delta\theta r^2, \quad r/r_{\text{ref}} = \sqrt{x/x_{\text{ref}}} \quad \text{when } A \propto x. \quad (1)$$

References are positive and shared across compared sectors. For $x = [25, 100]$, area encoding yields radius ratio 1:2 and area ratio 1:4; a radius linear in x yields area ratio 1:16. Linear radius is not intrinsically invalid: it conflicts with an area-based declaration about the same quantity.

For a fixed preprocessed three-axis signal $\mathbf{u}[n]$,

$$M_2 = \frac{1}{N} \sum_{n=0}^{N-1} \|\mathbf{u}[n]\|_2^2, \quad R = \sqrt{M_2}. \quad (2)$$

Scaling the signal by α scales RMS by $|\alpha|$ and mean square by α^2 . In particular,

$$r/r_{\text{ref}} = R/R_{\text{ref}} = \sqrt{M_2/M_{2,\text{ref}}} \implies A \propto M_2. \quad (3)$$

With calibrated acceleration, RMS has acceleration units and mean square squared acceleration units. Neither is mechanical power merely because it involves squaring.

Coverage and unobserved intervals

Let $m > 0$ of N sample positions be valid, $q = m/N$, and S the sum of squared magnitudes over valid positions only. Then

$$M_{2,\text{zero}} = S/N = q(S/m) = qM_{2,\text{valid}}. \quad (4)$$

For positive S , equal angles, and a common area scale,

$$A_{\text{zero}}/A_{\text{valid}} = q, \quad r_{\text{zero}}/r_{\text{valid}} = \sqrt{q}. \quad (5)$$

Here $N = 800$, $m = 200$, and $S = 20000$, so the feature values are 25 and 100. The unknown full-window quantity obeys, for $0 < q < 1$,

$$M_{2,\text{full}} = qM_{2,\text{valid}} + (1 - q)M_{2,\text{missing}}. \quad (6)$$

At $q = 1$, the valid and full summaries coincide. At $m = 0$, the valid-sample feature is undefined, not measured zero. A synthetic construction can have known latent values, but valid-sample normalization cannot recover unobserved measurements. Missing positions must be indexed out before squaring, rather than multiplied by a zero mask when they contain NaN.

Reviewing work and adopting the case

Review criteria, implementation conditions, and classroom distribution

A qualitative review checklist

Criterion	What the submitted work should make inspectable
Quantity and geometry	Names the encoded quantity and the geometric object actually compared; explains the ratio rather than supplies a number alone.
Feature semantics	Traces the feature to samples, a defined calculation and units; locates a repair in the computation, mapping, or caption.
Temporal support	Names the analysis window, duration, and aggregation; explains what the scalar cannot establish about burst timing or accumulation.
Observation support	States mask, denominator, shared scale, and treatment of no-observation bins; distinguishes a valid-sample summary from a full-window claim.
Interpretation and record	Retains, revises, or withdraws claims with reasons; records the supporting evidence, remaining uncertainty, and any model/source edits.

Use the checklist to review the reasoning recorded in each submission. A correct artifact can remain unchanged; a numerically correct chart with an unsupported caption remains incomplete.

Implementation conditions

Masks follow fixed preprocessing. Re-filtering a zero-filled signal changes the controlled comparison. The 800-point windows and two-hour labels are schematic, not a sampling-rate specification. Compare sector areas within panels: Fig. 3's quarter-sectors and daily 30-degree sectors have different area constants. A circular day alone is not evidence of circadian rhythm.

For a pooling extension, combine squared sums and valid counts before taking a square root. For sums [50,800] and counts [2,8], the pooled RMS is $\sqrt{85}$, not the mean of the two window RMS values [5,10]. This remains a valid-sample quantity, not automatically a full-interval statistic.

Distribution and provenance

Distribute cards in sequence and collect first responses before annotated reference figures. Keep the instructor guide and the reference page of the signal-audit extension separate from the initial learner handouts. Live GenAI is optional; keep the unedited output separate from teacher-constructed variants and record the model, date, prompt, checks, and edits. Use synthetic rather than personal recordings for the core activity.

The accompanying PDFs provide the learner worksheet, six staged release cards, and a two-page signal-audit extension. The extension addresses temporal support, local statistics, quality exclusions, and squared time accumulation. Its second page supplies reference decisions after learners record their predictions. The historical credit is on Card 0. The core activity is a paper-based audit using the supplied snippets and calculations.